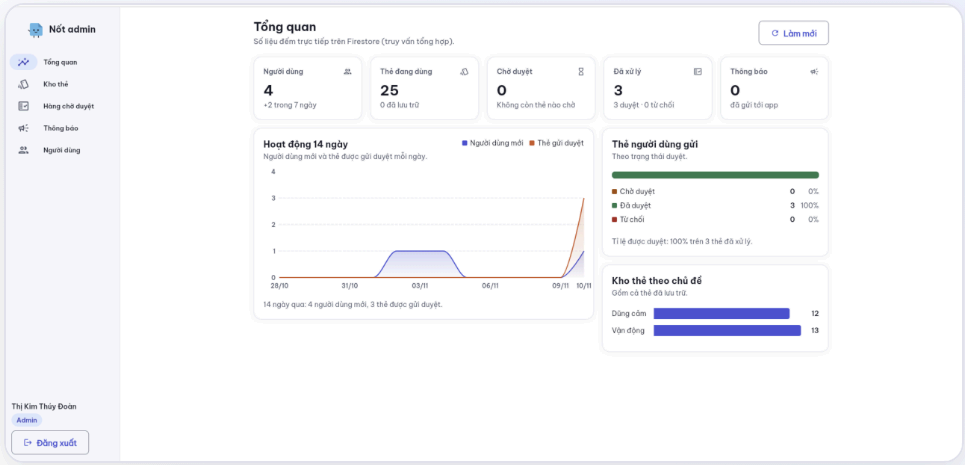
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Nốt

One positive note every day, now with staff, AI and a shared catalog

Đoàn Thị Kim Thúy · SE190992
Phùng Thế Quân · SE193519
Phan Đức Duy · SE193591
Phan Hữu Phước · SE161951





Nốt is a daily-wins app with a staff web admin, running on one Firebase project

The Lab 3 app became a two-client, three-role system: users log small wins on Android, staff curate the content they see.

USERS

Final-year students who feel behind; staff and admins who run the content

VALUE

Log a mood and one small win in under 30 s; get new challenge cards and news without updating the app

PLATFORM

Flutter Android app · Flutter Web admin on Hosting · Firestore, Rules, App Check, Gemini

WHAT THE FINAL ADDED ON TOP OF LAB 3

- Roles `user` / `staff` / `admin` and a web admin with 5 screens
- Cross-role flow: a user shares a card, staff approve it, the app updates in real time
- Nốt AI (Gemini via Firebase AI Logic): tell the day, get sorted notes to confirm
- Inbox for announcements, first-run onboarding, weekly card PDF saved on the device



App · Nốt AI

Live: prm393-not-lab3.web.app (app) · prm393-not-lab3.web.app/admin/ (web admin) · repo FU-PRM-Projects/Lab-03

Minh ends most evenings remembering what he did not finish

Lab 2 persona and the habit Nốt was designed around; Lab 3 shipped it for one person, alone.

PERSONA

Minh, 22, final-year student, worn out. Ends the evening looking at the list of unfinished tasks instead of what went well.

SUCCESS CRITERION (LAB 2)

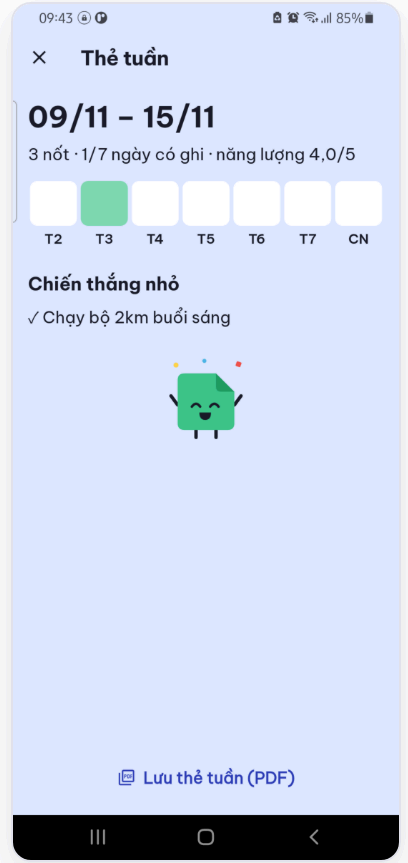
Log a mood and one small win in **under 30 seconds** after tapping the evening reminder.

HOW NỐT WORKED AFTER LAB 3

- One user, own data only (notes, habits, groups)
- Challenge cards: a fixed seed list in Firestore
- Own cards stayed private to their author

WHO ELSE IS INVOLVED NOW

- Staff who write and curate challenge cards
- An admin who decides who is staff
- Gemini, to cut typing in the evening



Weekly card, Lab 3 flow kept in the Final

Source: Lab 2 persona and success criterion (README, docs/design-traceability.md).

Three gaps kept Nốt from being a system: no curator, no channel, slow evenings

Each pain point is answered directly on the next slide.

PAIN POINT	CAUSE	IMPACT
Static challenge content	Cards were a seed list; changing one needed a developer and a script	Users run out of fresh cards; good ideas from users stay private
No way to reach users	No staff role, no announcements, push only from the Console	Nobody can tell users about new cards or events inside the app
Logging takes effort at night	Each note is typed and classified by hand (win, self-control, journal)	The 30 s goal can slip when the day had several moments to record
Lab 3 flow left unfinished	Weekly card PDF went to Cloud Storage, which needs the paid plan	Export never worked on the real project

Evidence: docs/gap-review.md (Lab 2 / Lab 3 gaps before the Final). No user survey was run; impacts are observed in the Lab 3 build.

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One Firebase system: a staff web admin, a review flow, and Gemini that sorts the day

Every pain point maps to a feature that is live and demonstrated.

PAIN POINT	SOLUTION	EXPECTED VALUE
Static content	Web admin catalog + users share cards, staff approve	Catalog grows without a release
No channel	Announcements + personal inbox (Firestore, real time)	News lands in the app in the same minute
Slow evenings	Nốt AI: tell the day once, confirm sorted notes	Several notes from one paragraph
Broken PDF	Build and save the PDF on the device, metadata in Firestore	Works on the free plan, offline

IN SCOPE (MVP)

- WF-01 Log the day with Nốt AI
- WF-02 Publish a user card to the catalog
- WF-03 Announce to all users
- Onboarding, weekly PDF, roles, App Check

OUT OF SCOPE

- Cloud Functions, Cloud Storage (brief + paid plan)
- AI level 3 "Hỏi Nốt" (bonus, not built)
- iOS build

Three human roles and two external services, enforced by Security Rules

The role lives in users/{uid}.role; only an admin changes it, and never their own.

ACTOR	CLIENT	WHO	MAIN PERMISSIONS
User	Android app	Student using Nốt every evening	Own notes and data; share own cards (pending only); read catalog, announcements, own inbox
Staff	Web admin	Content curator	Create, edit, archive catalog cards; approve / reject with reason; post announcements
Admin	Web admin	Team lead	Everything staff can do + change other users' roles
Gemini	External	Firebase AI Logic, Gemini Developer API	Receives the scrubbed paragraph only; returns JSON suggestions
Firebase	Platform	Auth, Firestore, App Check, RC, FCM…	Rules decide every read and write; no custom server

First admin: set in the Firebase Console (Thị Kim Thúy Đoàn). Rules tests R-19…R-28 cover every role and collection.

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Three core workflows carry the demo; everything else supports them

Each workflow has an actor, a trigger and a business result, not a list of buttons.

WF-01

Log the day with Nốt AI

Actor: User
Trigger: Evening, user taps Ghi nốt → Kể tự do
Result: Notes saved after the user confirms

Consent · Gemini JSON · edit · fallback

WF-02

Publish a user card to the shared catalog

Actor: User + Staff
Trigger: User shares an own card
Result: Card in catalog, app updates, inbox message

Submit · review · approve / reject + reason

WF-03

Announce to all users

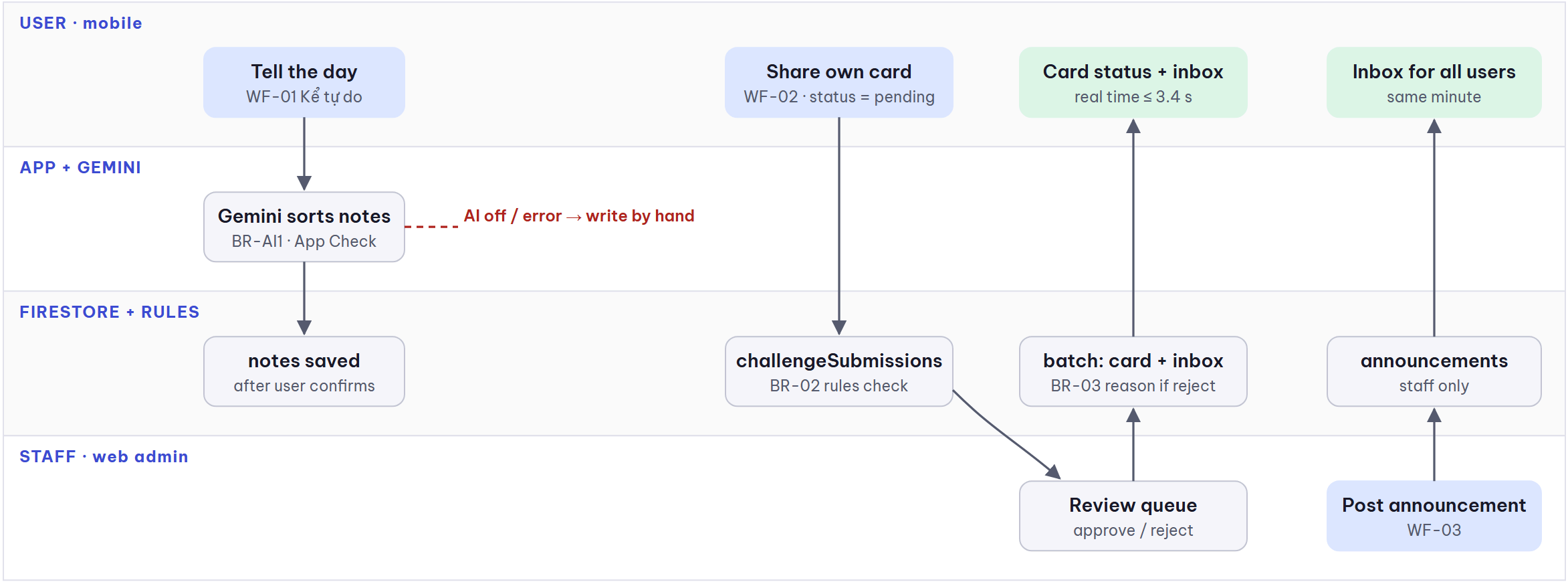
Actor: Staff
Trigger: Staff has news for everyone
Result: Message in every user inbox

Form · validation · real-time inbox

Other features: onboarding & starter plan · weekly card PDF · streaks & journey · self-control & breathing · groups · Discord link · dashboard · roles.

One swimlane view: who acts, where the data lands, where it branches

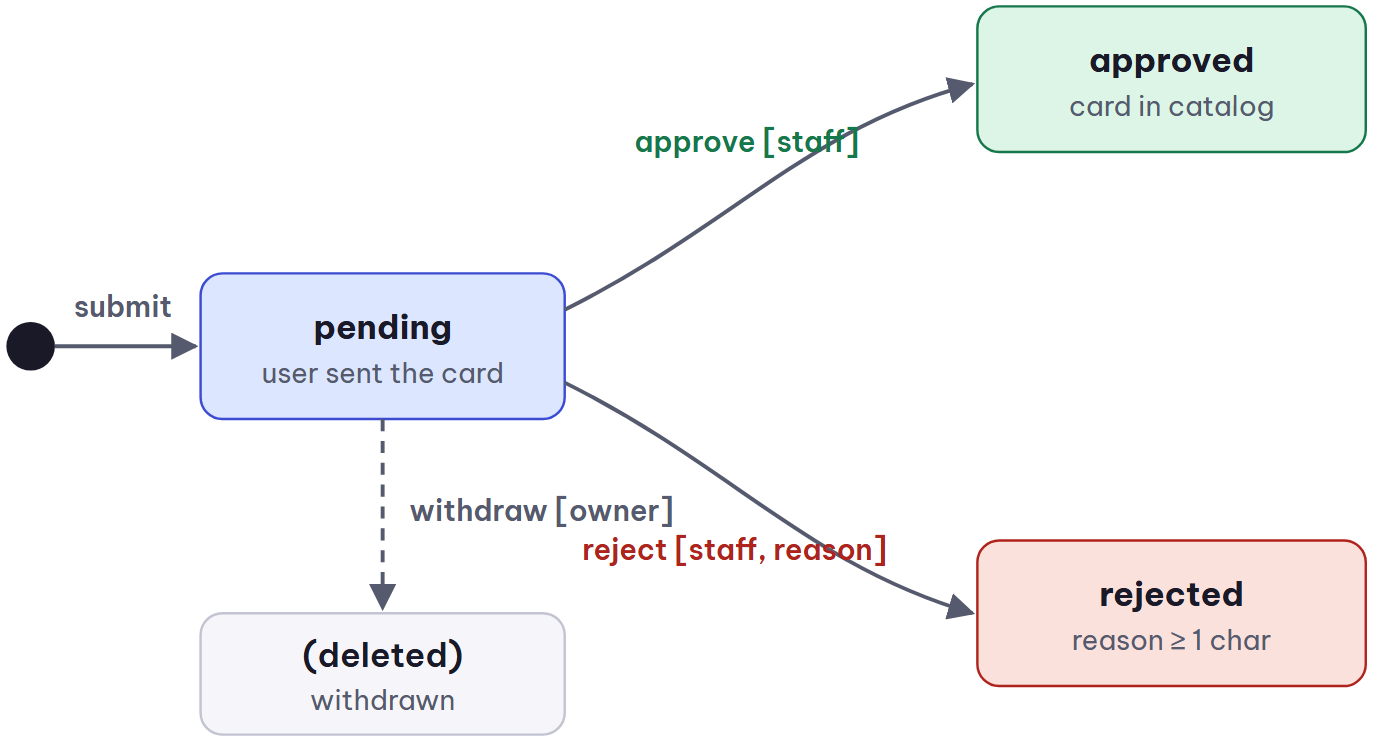
The three workflows share one Firestore; rules are the decision points.



BR-AI1: AI only when enabled, consented and App Check passes · BR-02: user creates pending only · BR-03: reject needs a reason.

A shared card moves pending → approved or rejected, and nowhere else

Entity: challengeSubmissions/{id}. Every transition and every forbidden one has a rules test.



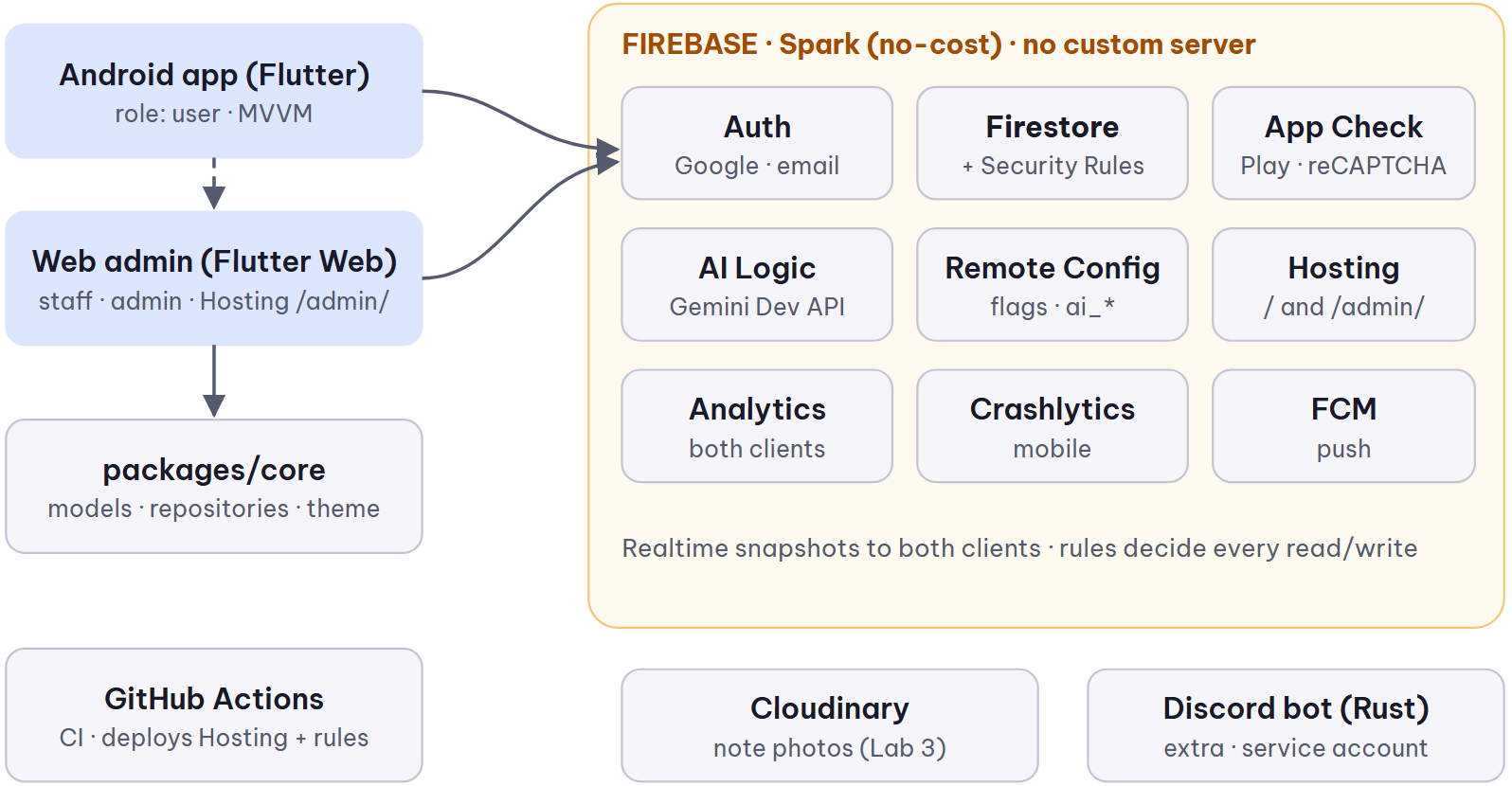
Forbidden (rules deny)

user → approved
approved → pending
reject without reason
withdraw after review

Tests: R-20 create pending only · R-21 read own or staff · R-22 staff review, reason on reject · R-23 withdraw while pending. 28/28 rules tests pass.

Two Flutter clients, one shared package, Firebase as the only backend

Decision: no custom server. Security Rules + App Check are the backend's access layer.



Deploy: GitHub Actions builds both clients, deploys Hosting (/ and /admin/) and Firestore rules + indexes on merge to main.

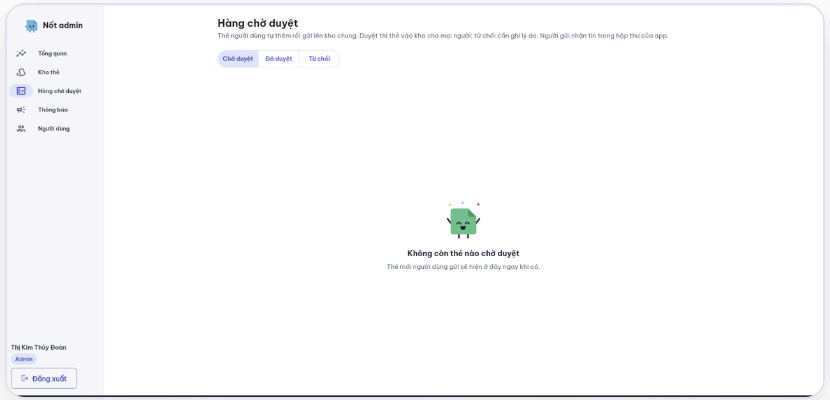
Every technology in the stack has one job

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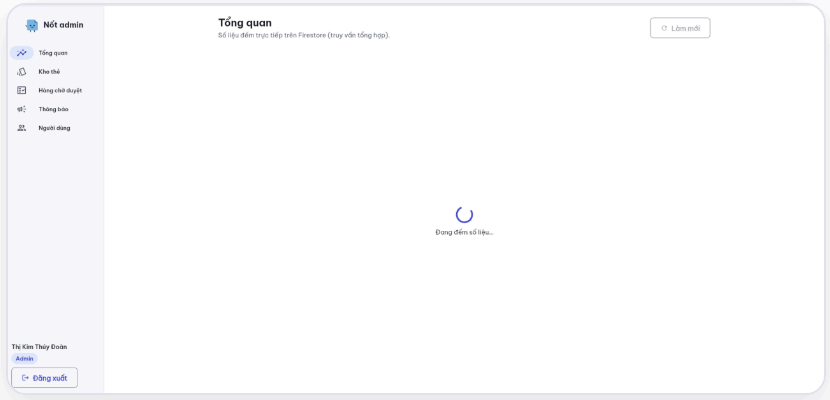
GROUP	TECHNOLOGY	USED FOR
Mobile	Flutter · Riverpod 3 · go_router · fl_chart · pdf · share_plus	Android app, MVVM, charts, weekly PDF on device
Web	Flutter Web · same Riverpod/MVVM · google_fonts	Web admin on Firebase Hosting
Shared	`packages/core` (Dart)	Models, Firestore repositories, design tokens
Backend	Firestore · Security Rules · Auth · Remote Config · FCM	Data, permissions, flags, push
AI	Firebase AI Logic (`firebase_ai`) · Gemini Developer API	Nốt AI: sort a paragraph into notes (JSON schema)
Security	App Check: Play Integrity / debug · reCAPTCHA Enterprise	Only our apps reach Firestore and Gemini
Quality	flutter_test · Firebase Emulator + rules-unit-testing · ARTEMIS · CodeRabbit	Unit/widget, rules, E2E on device, AI review
DevOps	GitHub Actions · Firebase Hosting preview channels	CI, PR previews, deploy on merge

One design system across phone and web, with every UI state designed

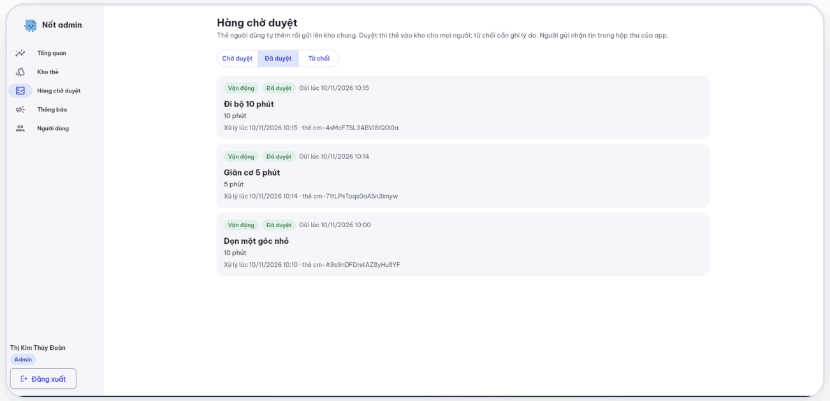
Lab 2 tokens live in packages/core; the web admin reuses them in a wide layout.



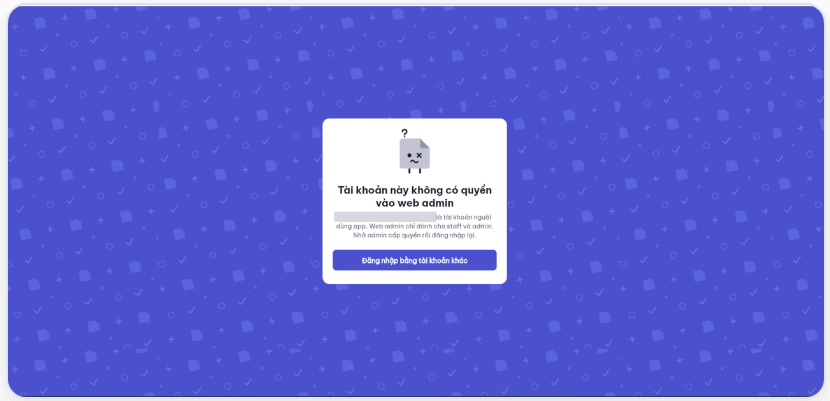
Web · empty: Hàng chờ duyệt



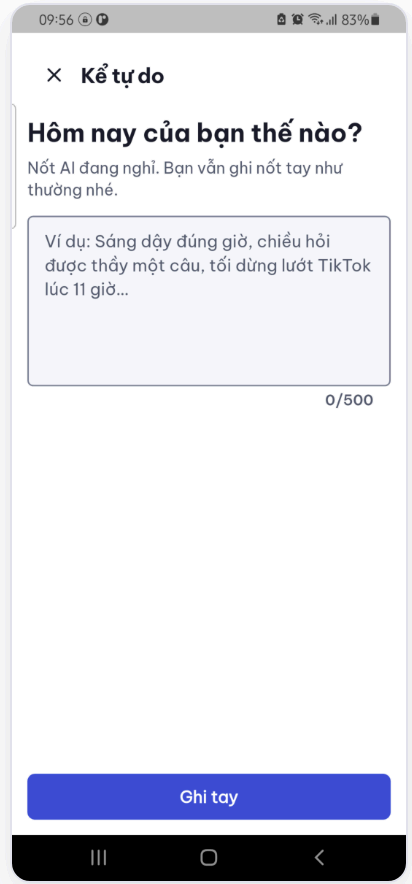
Web · loading: Tổng quan



Web · success: approved cards



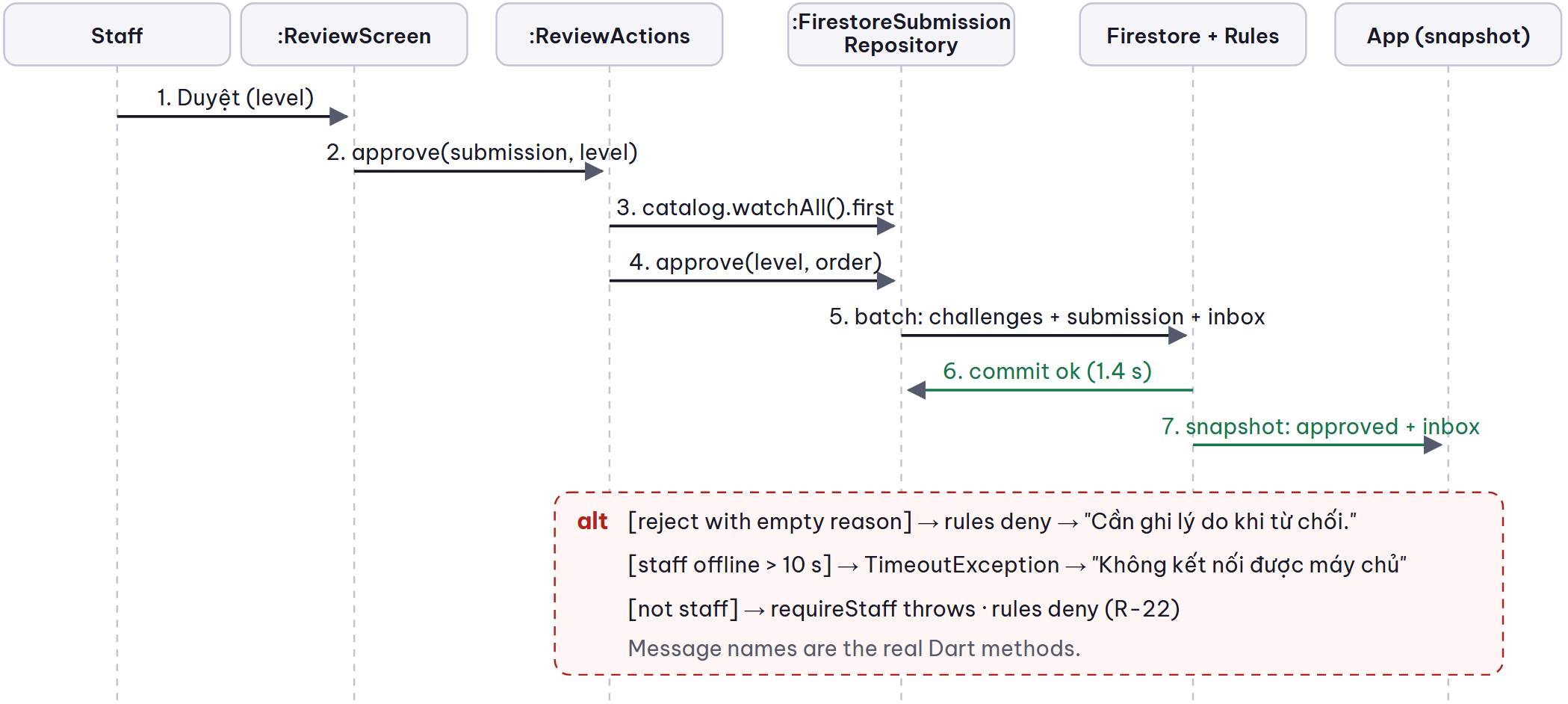
Web · role rejected (mascot, clear message)



App · AI off state

States: empty (review queue), loading (dashboard), error (shared MessageView.error on 5 screens), success (approved list).

Approving a card is one batch write; rules guard it, snapshots carry it back



MVVM in both clients: screens never touch Firebase directly

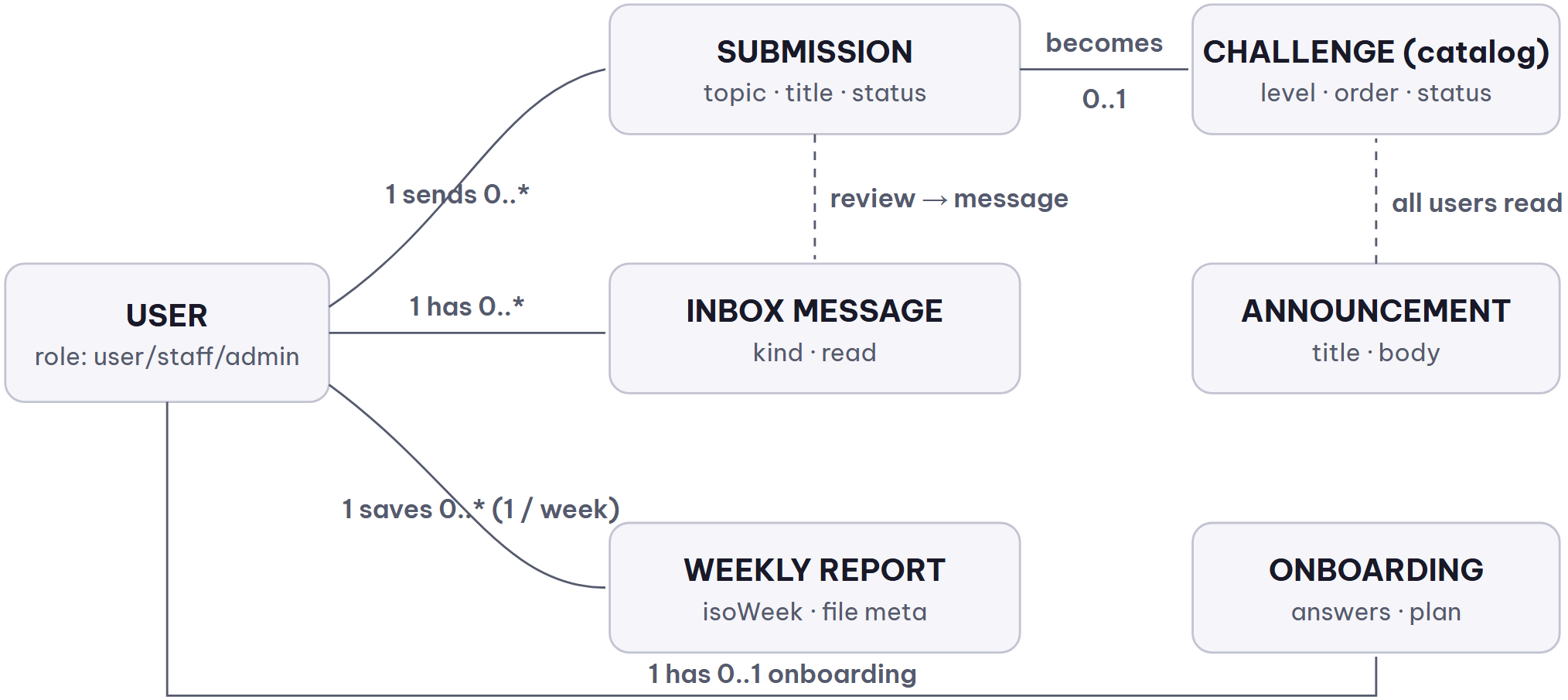
Business rules sit in the repository + Security Rules, not in widgets.

LAYER	WF-01 NỐT AI	WF-02 SHARE & REVIEW	WF-03 ANNOUNCE
Screen	`FreeTellScreen`	`ChallengesScreen` · `ReviewScreen`	`AnnouncementsScreen` · `NotificationInboxSheet`
ViewModel	`FreeTellViewModel`	`CommunityActions` · `ReviewActions`	`AnnouncementActions` · `communityInboxProvider`
Service	`AiService`	`SubmissionRepository` · `CatalogRepository`	`AnnouncementRepository` · `InboxRepository`
Implementation	`FirebaseAiService`	`FirestoreSubmissionRepository`	`FirestoreAnnouncementRepository`
Model	`AiSortResult` · `AiNoteSuggestion`	`ChallengeSubmission` · `CatalogCard`	`Announcement` · `InboxMessage`

main.dart wires each provider to its Firebase implementation; tests swap in fakes (290 app · 10 web admin · 5 core tests).

The Final added six entities around the user

Lab 3 entities (notes, habits, groups, progress) stay private to their owner.



Firestore collections, with constraints written as Security Rules

COLLECTION	KEY FIELDS	CONSTRAINTS (RULES)
`users/{uid}`	role, displayName, email	self-create as user; only admin changes role, never own
`challenges/{id}`	topic, level 1–10, order, title ≤ 80, status	staff only; updatedBy = auth.uid; archive, no delete
`challengeSubmissions/{id}`	uid, topic, title, minutes, status, reason	create pending; staff review; reason on reject
`users/{uid}/inbox/{id}`	kind, title, body, read	staff create; owner reads, may only flip read
`announcements/{id}`	title ≤ 80, body ≤ 1000, authorUid	staff create/delete; every signed-in user reads
`users/{uid}/reports/{week}`	isoWeek, fileName, sizeBytes, noteCount	owner only; id = week; savedAt = server time
`users/{uid}/private/onboarding`	answers, profileLabel, planSource	owner only; enums and sizes checked

Indexes: submissions (uid, createdAt↓), (status, createdAt↓) · notes (groupId, createdAt↓) · full table: docs/data-model.md

Each workflow traces from screen to rule to test

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WORKFLOW	SCREENS	CLASS / SERVICE	COLLECTION	TESTS
WF-01 Nốt AI	S03b Kể tự do	FreeTellViewModel · AiService	notes	free_tell_flow_test · real-device run · 3 real Gemini examples
WF-02 Share & review	S07 · Review queue · Inbox	CommunityActions · ReviewActions	challengeSubmissions · challenges · inbox	R-20…R-24 · session_test (approve) · D-1… D-3
WF-03 Announce	Announcements · Inbox	AnnouncementActions	announcements	R-25 · notification_inbox_test · web checklist #9–11
Weekly PDF	S11b Thẻ tuần	WeeklyReportViewModel	reports	R-28 · weekly_report_test · real-device run
Onboarding	S13a–g	OnboardingViewModel	private/onboarding	R-27 · onboarding_flow_test
Roles	Sign-in · Users	sessionProvider · UserActions	users	R-01, R-26 · session_test (reject user)

Full matrix: docs/design-traceability.md · rules: firebase/tests/firestore.rules.test.js · E2E: e2e_test/

The demo is one evening: Minh logs his day, shares a card, staff publish it and announce

The result of each workflow is the input of the next.

01 · USER
Kể tự do: one paragraph → 3 notes saved

→

02 · USER → STAFF
Share "Đi bộ 10 phút" → staff approves → app shows it

→

03 · STAFF → USERS
"Bạn đã vận động chưa?" → every inbox

RESULT WF-01
Jar 11 → 14 notes, streak starts

RESULT WF-02
Card "Người dùng đề xuất" in catalog; inbox message

RESULT WF-03
Announcement in app inbox at 10:56, same minute

EXCEPTION PATHS SHOWN LIVE

- Remote Config `ai_enabled = false` → "Nốt AI đang nghỉ", user writes by hand
- Reject without reason → blocked · non-staff account → rejected at sign-in

Prepared: admin account (web) · user account on Galaxy A70 with App Check debug token · Remote Config ai_enabled ready to toggle · backup video.

WF-01 Log the day with Nốt AI: one paragraph becomes three editable notes

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TRIGGER

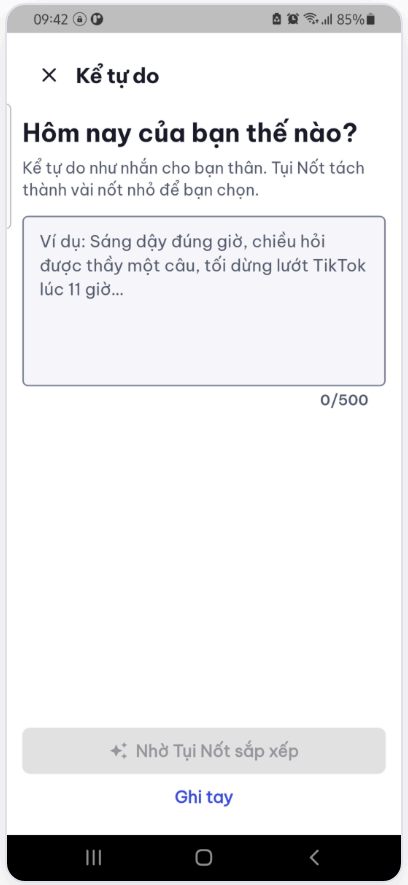
Evening; user taps Ghi nốt → Kể tự do

STEPS

- First use: consent sheet (what is / is not sent)
- Paragraph scrubbed (email, phone), ≤ 500 chars
- Gemini returns notes + mood as JSON
- Labelled "Gợi ý từ Nốt AI", editable, untickable

EXCEPTION

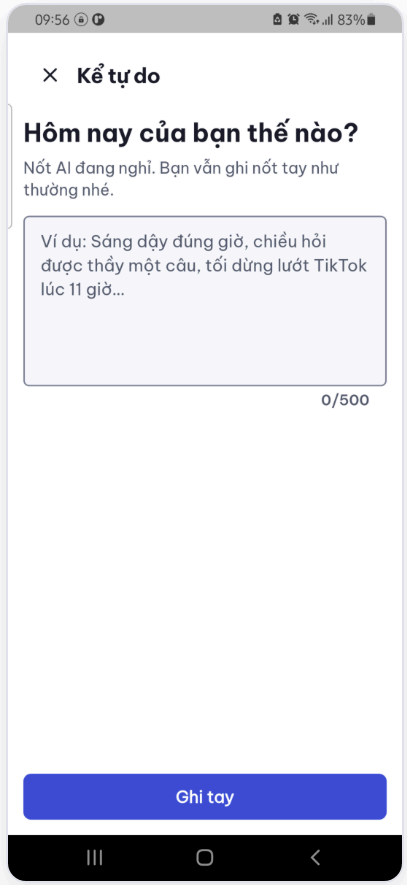
AI off / quota / network / App Check → message + "Ghi tay"; the paragraph is kept



Consent



Suggestions



ai_enabled = false

Run 10/11/2026 on Galaxy A70 against the real project · model gemini-flash-lite-latest (~2 s) · AI level 2: JSON schema → AiSortResult, confirmed by the user.

WF-02 Publish a card: staff approve on the web, the phone updates within seconds

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STATE

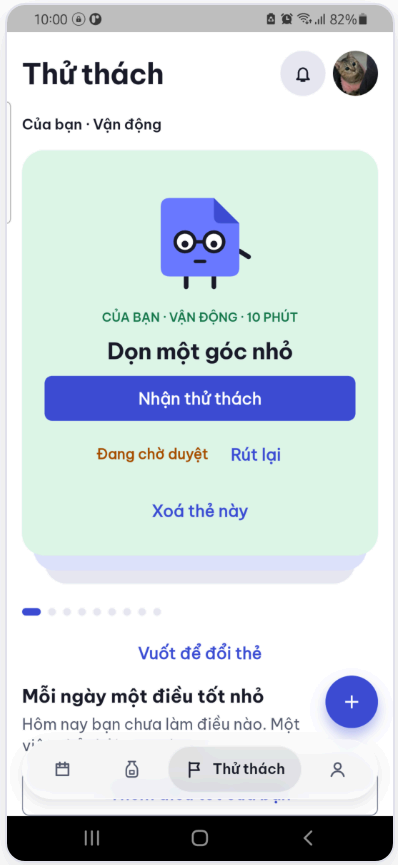
pending → **approved** (or rejected + reason)

STEPS

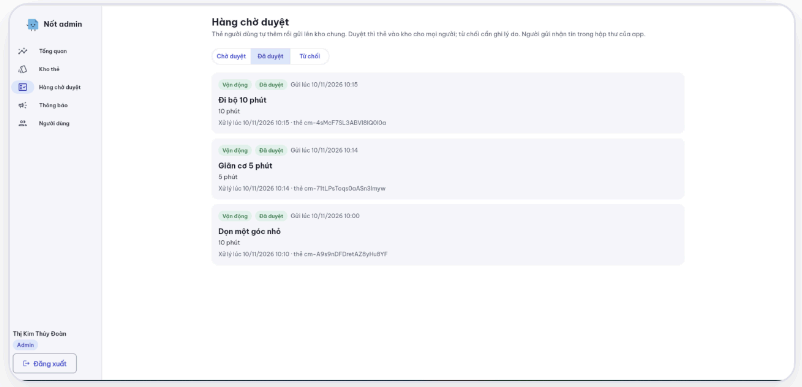
- User: Gửi cho mọi người → pending
- Staff: Hàng chờ duyệt → Duyệt, pick level
- Batch: catalog card + status + inbox
- App: "Đã vào kho chung ✓", bell badge

EXCEPTION

Reject needs a reason (rules + UI) · user cannot approve (R-22)



Đang chờ duyệt



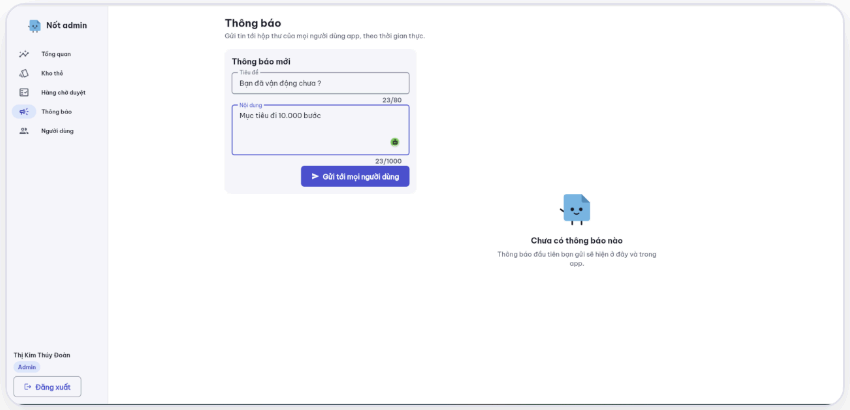
Web: approved



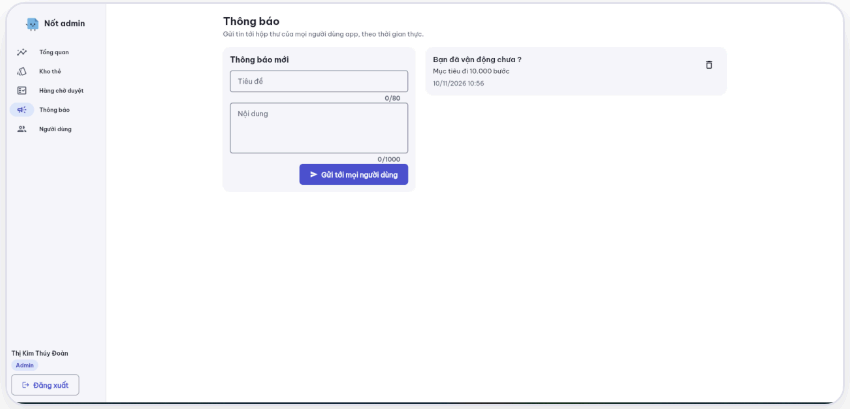
App: inbox

Server write 1.4 s after Duyệt in all runs · phone update ≤ 3.4 s (D-2), ≤ 2.3 s (D-3); D-1 22 s after the listener sat idle · docs/cross-role-flow.md

WF-03 Announce: staff post once, every user sees it in the inbox in real time



1 · Compose (list still empty)



2 · Sent at 10:56, form cleared



3 · App inbox, 10:56

TRIGGER

Staff has news for every user

RULES

Staff only · authorUid = self · title ≤ 80 · body 1–1000

EXCEPTION

Empty body or non-staff → rules deny (R-25)

Rules R-25: only staff create, authorUid = self, non-empty body; every signed-in user reads · web checklist #9–11.

Rules, code and the deployed system are each tested, with evidence in the repo

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28/28
Security Rules tests

290
Flutter app tests

10 + 5
web admin + core tests

41/42
ARTEMIS runs (Lab 3)

LEVEL	EVIDENCE	RESULT
Rules	Emulator, allowed + denied per role per collection	28 / 28
Unit / widget	ViewModels, repositories, screens at 360 dp	all pass
Cross-role	3 runs on the deployed system, timed	3 / 3
Web checklist	13 screenshots: actions, rejection, empty, loading	done
Real device	Onboarding, Nốt AI, PDF, AI off on Galaxy A70	pass
E2E (ARTEMIS)	Lab 3 flows 3× each; Final flows written, not run yet	partial

CI on every PR: format, analyze, all test suites, rules in the Emulator, web build, Android build, Hosting preview.

Every brief requirement maps to a live feature and a piece of evidence

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BRIEF REQUIREMENT	IMPLEMENTED	EVIDENCE
Web admin: 4 screens + announcement form, deployed	Dashboard, catalog, review, announcements, users	`evidence/web-admin/` (13 shots)
Role-based rules, deny by default, tested	user / staff / admin, R-01…R-28	`evidence/firestore/rules-tests.txt`
Cross-role flow, real time, documented	Share card → approve → app + inbox	`docs/cross-role-flow.md`, `evidence/cross-role/`
AI level 2 with RC, App Check, fallback	Kể tự do on Gemini, `ai_*` flags, enforced App Check	`docs/ai-feature.md`, `evidence/mobile-final/`
App Check on Firestore and AI Logic	Web: reCAPTCHA Enterprise; Android: Play Integrity + debug	`evidence/app-check/` (43 verified)
Lab 3 PDF saved & shared on device	pdf + share_plus, metadata in Firestore	15.4 KB file pulled from the phone
Mobile data on Firestore with inbox	Announcements + personal inbox, real time	web checklist #11

What is not done yet, and what it costs

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LIMITATION	IMPACT	PRIORITY	NEXT STEP
ARTEMIS not run for the Final flows	Onboarding, Nốt AI, cross-role mobile steps lack 3× E2E evidence	High	Run on the Galaxy A70, 3× each
App Check on Firestore only monitored	Unverified clients can still reach Firestore (rules still apply)	Medium	Enforce once almost all traffic is verified
Lab 2 flows A3 (practice chat) and P4 (resume draft) not built	Two designed flows missing	Medium	Build A3 on the same AiService
D-1 took 22 s on an idle phone	Real time not guaranteed after long idle	Low	Measure more runs; re-listen on resume
Release builds sideloaded fail Play Integrity	AI blocked outside Play / debug builds	Low	Demo on debug build; Play internal track

Running the system for real found bugs the tests did not

Three bugs surfaced only against the real project; all three are fixed, the first two with regression tests.

APPROVE TIMED OUT

Riverpod paused the unwatched catalog provider; approval waited 10 s and failed. Found while timing D-1. Fixed: read the repository once (PR #70).

DASHBOARD SPUN OFFLINE

`count()` waits for the server forever. Found while shooting the error state. Fixed: 15 s timeout → error state (PR #71).

GEMINI MODEL RETIRED

`gemini-2.5-flash` returned 404 for new projects. Fixed: default `gemini-flash-lite-latest`, model stays in Remote Config.

RULES ARE THE BACKEND

Without a server, every business rule is a rules condition, so rules need their own test suite.

READ THE BRIEF TWICE

We removed the PDF for "no Cloud Storage", then re-read: the brief wanted it on the device. Restored.

AI NEEDS A PLAN B

App Check, quota and model changes all broke AI at some point; the hand-written path kept the flow alive.

Nốt is now a working multi-role system, and we can show every claim

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1 · PROBLEM SOLVED

Content is curated by staff and grows from users' own cards; news reaches every user in the same minute.

2 · WORKING PRODUCT

WF-01, WF-02, WF-03 run live on Android and the deployed web admin, on the free plan.

3 · VERIFIED

28/28 rules tests, 300+ code tests in CI, 3 timed cross-role runs, real-device AI and PDF evidence.

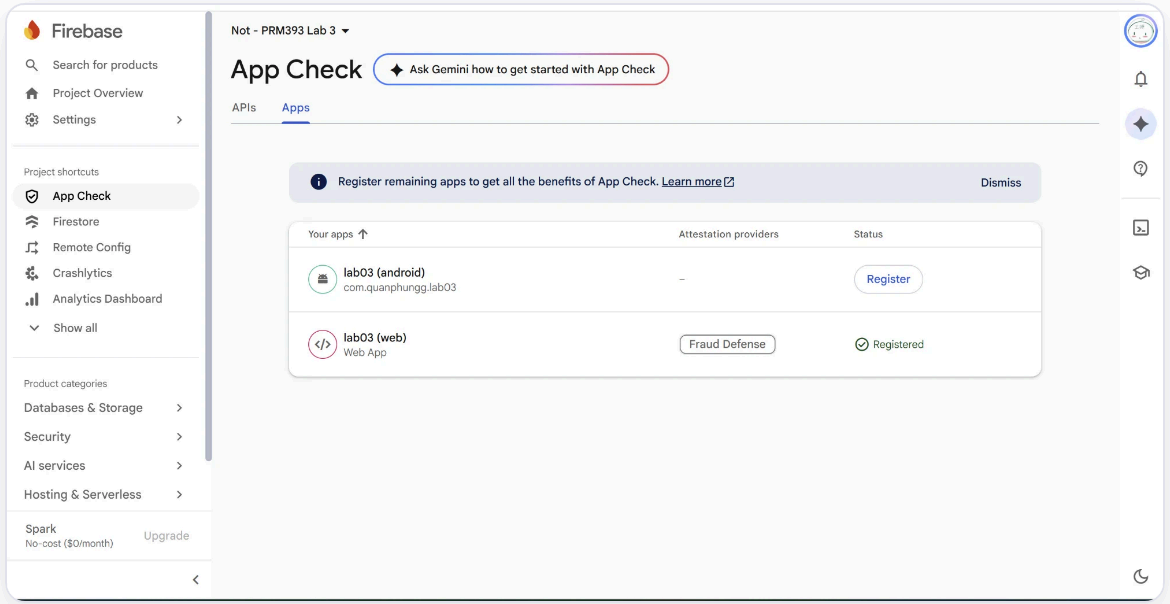
Demo: prm393-not-lab3.web.app · Web admin: prm393-not-lab3.web.app/admin/ · Repo: github.com/FU-PRM-Projects/Lab-03

Three real Gemini inputs and outputs, and one handled failure

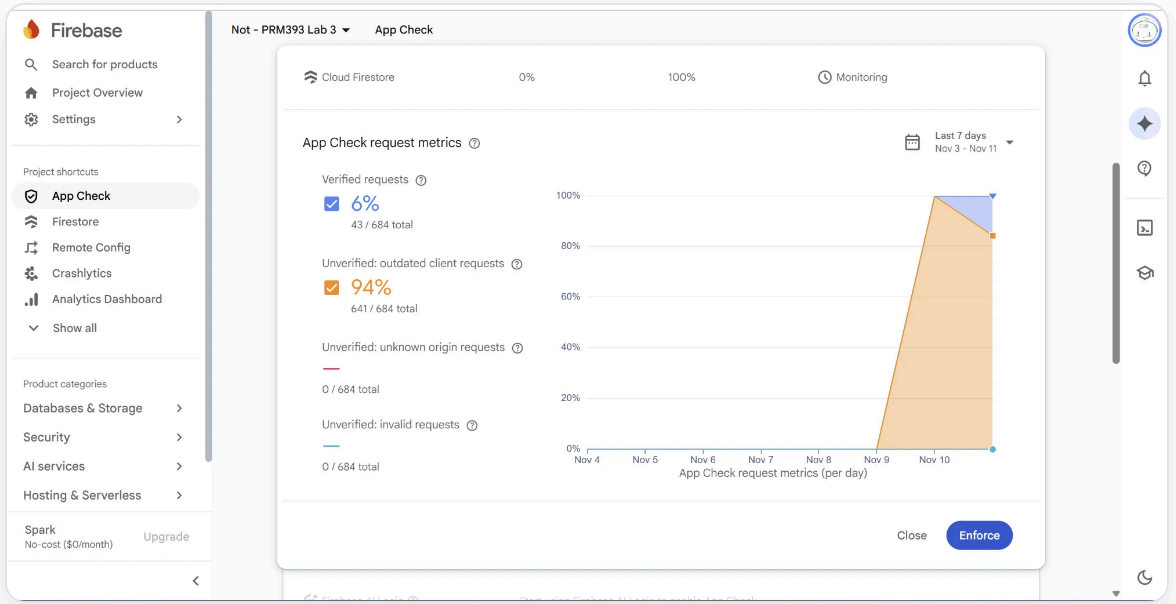
INPUT (KỂ TỰ DO)	GEMINI OUTPUT
Ran 2 km, ate a snack I meant to skip, calmed down without blaming myself	chien_thang "Chạy bộ 2km quanh hồ" · nhat_ky "Lỡ ăn thêm gói snack…" · tu_chu "Không tự trách và bình tĩnh lại" · mood binh_thuong
Asked a stranger in the library where to charge my phone; wrote a few lines about the rainy afternoon	chien_thang "Lấy hết can đảm hỏi mượn chỗ sạc pin" · nhat_ky "Ghi lại vài dòng cảm xúc buổi chiều mưa" · mood vui
Tired for days, only tidied my desk today	chien_thang "Mình đã dọn dẹp được cái bàn học" · 2× nhat_ky · mood met · needs_support false
Handled failure	404 model retired / 401 missing App Check → `ai_error` logged, paragraph kept, "Ghi tay" offered; fixed by Remote Config model name

Inputs are scrubbed (email, phone) and capped at 500 characters; no names, emails or photos are sent.

Both apps are registered; Firestore shows verified traffic from the web admin



Apps registered



Firestore: 43 verified, 0 invalid

Web: reCAPTCHA Enterprise (Fraud Defense) · Android: Play Integrity + debug tokens · Firebase AI Logic: enforced.